Alignment-Aware Neural Architectures (AANA): Constraint-Coherent Learning via Verifier-Grounded Correction

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Abstract

Modern neural systems are trained to optimize proxy objectives that only partially capture real-world correctness, safety, and coherence. This gap leads to failure modes such as hallucination, brittleness, and unsafe outputs. We introduce Alignment-Aware Neural Architectures (AANA), a framework that augments neural models with constraint-aware verification, misclassification yield estimation, and iterative correction dynamics. We formalize alignment as membership within a feasible constraint region and model inference as a dynamical system under constraint pressure. We derive conditions for contraction, stability, and efficiency improvement; specify a practical architecture and training procedure for verifier-grounded correction; and present an experimental evaluation protocol together with simulated comparative results. The framework suggests that explicitly modeling constraint satisfaction and correction can improve robustness and reduce downstream error propagation relative to standard generation pipelines.

1 Introduction

Large-scale neural networks achieve strong performance across tasks by optimizing proxy objectives such as likelihood or reward [3, 6]. However, these objectives do not fully capture real-world constraints, leading to well-documented failure modes including hallucinations, miscalibration, brittle reasoning, and unsafe outputs [1, 5]. In many deployments, systems improve on benchmark metrics while diverging from the deeper requirements that determine whether their outputs are actually usable.

We propose that these failures arise from a structural mismatch:

$$\nabla \mathcal{L}_{task} \nparallel \nabla A$$

where A denotes alignment with real-world constraints. The core issue is not merely that current objectives are imperfect, but that standard neural pipelines lack a structural mechanism for evaluating whether generated outputs remain within a feasible region once proxy optimization pressure increases.

This paper introduces Alignment-Aware Neural Architectures (AANA), which incorporate constraint evaluation and correction directly into the inference loop. The central claim is that neural systems should be designed not only to produce correct outputs, but to remain *correctable under pressure*. This reframes alignment as a dynamic maintenance problem rather than a static property of training.

Our contributions are fourfold:

1. We formalize alignment as membership within a feasible constraint region composed of physical, biological, and constructed constraints.

2. We model neural inference as a dynamical system under optimization pressure, misclassification, and correction.
3. We specify a practical architecture combining a base generator, verifier stack, grounding module, correction policy, and alignment gate.
4. We derive theoretical and systems-level results showing when verifier-grounded correction improves stability and alignment efficiency.

2 Problem Formulation

Definition 1 (Constraint Space)**.**

$$\mathcal{R} = (K_P, K_B, K_C)$$

where K_P denotes physical or factual constraints, K_B denotes biological or human-impact constraints, and K_C denotes constructed or task-level constraints.

Definition 2 (Feasible Region)**.**

$$\mathcal{F} = K_P \cap K_B \cap K_C$$

The feasible region is the set of outputs that satisfy all relevant constraint classes simultaneously.

Definition 3 (Alignment)**.** *For input x and candidate output y,*

$$A(x, y) = \mathbb{P}(y \in \mathcal{F})$$

and corresponding misalignment is

$$M(x, y) = 1 - A(x, y).$$

The underlying optimization problem is:

$$\min \mathcal{L}_{task} \quad \text{subject to} \quad y \in \mathcal{F}.$$

In standard neural systems, the constraint is implicit or absent. AANA makes it explicit.

2.1 Interpreting the Constraint Classes

The partition into K_P, K_B, K_C is not meant to imply a metaphysically rigid decomposition of reality. Rather, it is an engineering abstraction that distinguishes three practically relevant kinds of failure:

- **Physical / factual failures** (K_P): unsupported claims, contradictions to evidence, numerically invalid outputs, impossible actions.
- **Biological / human-impact failures** (K_B): manipulative responses, harmful instructions, cognitive overload, poor calibration to user context.
- **Constructed / task failures** (K_C): failing the requested format, violating role or policy constraints, producing incoherent plans, or ignoring instructions.

This decomposition matters because different corrective mechanisms address different failure types. Retrieval improves K_P grounding. Safety filtering improves K_B. Format control improves K_C.

3 Alignment as a Dynamical System

We model alignment evolution at the system level as:

$$\frac{dA}{dt} = -\pi\epsilon(1-\gamma) + C_{AI} - \Phi$$

where:

- π is optimization pressure,
- ϵ is misclassification rate,
- γ is constraint visibility or feedback integrity,
- C_{AI} is correction capacity,
- Φ is drift or irrecoverable loss.

This formulation captures a basic intuition: alignment degrades when optimization pressure acts on errors under weak feedback; it improves when correction capacity exceeds divergence pressure.

3.1 Interpretation

The dynamical equation should be read as a coarse systems model rather than a literal physical law. It is useful because it expresses four design levers:

1. reduce misclassification ϵ,
2. improve feedback visibility γ,
3. increase correction capacity C_{AI},
4. reduce environmental or representational drift Φ.

AANA primarily targets the second and third terms while indirectly reducing the first.

4 Architecture

We define an AANA system as:

$$\mathcal{S} = (f_\theta, E_\phi, R, \Pi_\psi, G)$$

where:

- f_θ: base generator,
- E_ϕ: verifier stack,
- R: retrieval / grounding module,
- Π_ψ: correction policy,
- G: alignment gate.

4.1 System Loop

The core inference loop is:

$$
\begin{aligned}
y_0 &\leftarrow f_\theta(x) && (1)\\
s_t &\leftarrow E_\phi(x, y_t, e_t, m_t) && (2)\\
e_t &\leftarrow R(x, y_t) && (3)\\
a_t &\leftarrow \Pi_\psi(x, y_t, s_t, e_t, m_t) && (4)\\
y_{t+1} &\leftarrow T_{a_t}(x, y_t, s_t, e_t) && (5)
\end{aligned}
$$

The action space includes:

$$\mathcal{A} = \{\texttt{accept}, \texttt{revise}, \texttt{retrieve}, \texttt{ask}, \texttt{refuse}, \texttt{defer}\}.$$

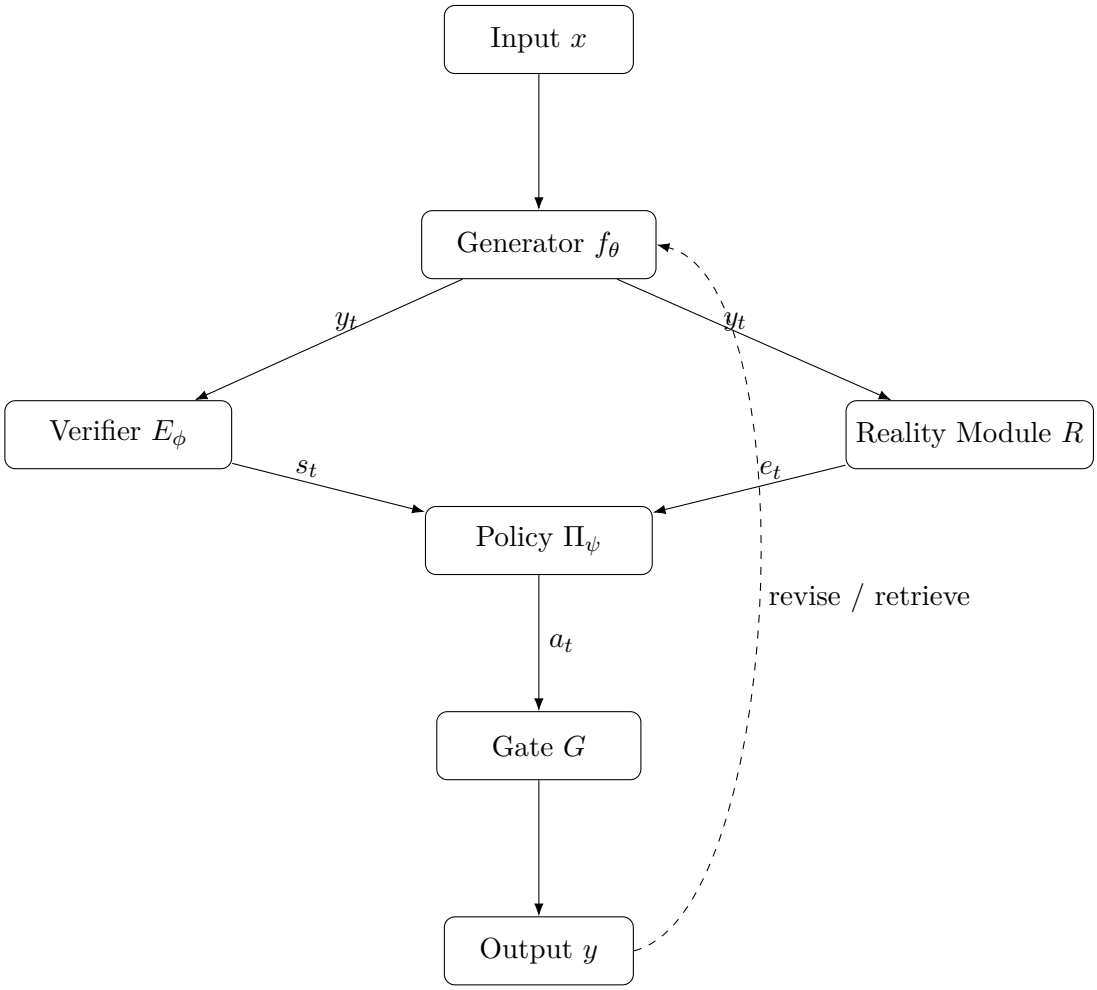

Figure 1: Alignment-Aware Neural Architecture loop. The generator proposes an output; the verifier and grounding module evaluate it; the policy selects a corrective action; and the gate decides whether to emit, revise, or withhold the output.

5 Main Results

We summarize the key theoretical and systems contributions of AANA:

- AANA introduces a correction operator that can induce contraction in misalignment space.
- Alignment dynamics can be modeled as a control system with explicit stability conditions.
- Misclassification yield β governs the rate of alignment degradation under proxy optimization.
- Alignment efficiency improves when correction reduces downstream error propagation.

6 Verifier System

Each constraint domain is approximated via:

$$v_k(x, y, e, m) \in [0, 1], \quad k \in \{P, B, C, F\}$$

where F denotes feedback integrity or confidence justification.

The aggregate verifier score is:

$$\hat{A}(x, y) = \sum_k w_k v_k$$

with $\sum_k w_k = 1$ and $w_k \geq 0$.

6.1 Implementation

Verifiers are implemented as lightweight transformer classifiers or cross-encoders trained on domain-specific supervision signals:

- v_P: factual correctness, contradiction to retrieved evidence, numerical validity,
- v_B: safety labels, manipulation risk, human-impact annotations,
- v_C: instruction adherence, formatting, plan coherence,
- v_F: calibration targets, uncertainty-quality alignment, verifier agreement.

A practical configuration uses small models relative to the base generator so that verification is cheaper than full regeneration.

6.2 Training Objective

The verifier stack is trained with:

$$\mathcal{L}_{verifier} = \sum_k \mathbb{E}[\ell_k(v_k, z_k)]$$

where z_k are domain-specific labels.

6.3 Calibration

For the correction loop to be meaningful, verifier scores must be calibrated enough to track actual constraint satisfaction. We therefore assume a bounded calibration error:

$$|\hat{A}(y) - A(y)| \leq \epsilon.$$

This assumption is used in the contraction proof below.

7 Misclassification Yield

We define misclassification yield as:

$$\beta = \frac{\tau_{lag} \cdot \sigma \cdot \mu}{1 + \xi \cdot \rho}$$

where:

- τ_{lag}: lag between error and observable consequence,
- σ: spread or diffusion of downstream cost,
- μ: opacity of true error under available supervision,
- ξ: irreversibility of downstream action,
- ρ: system awareness of irreversibility.

Intuitively, β measures how rewarding it is for the system to treat a real constraint as if it were negotiable. High-β domains are settings where proxy optimization can drift far from reality before being penalized.

7.1 Using β

We use β to scale constraint pressure:

$$\lambda_k = \lambda_k^{base}(1 + c\beta)$$

so that domains with high misclassification yield receive stronger verifier and correction pressure during training or inference.

8 Correction Policy

The correction policy decides what to do when the candidate output is insufficiently aligned. It maps the current state to an action:

$$a_t \sim \Pi_\psi(x, y_t, s_t, e_t, m_t).$$

8.1 State Representation

A simple policy state is:

$$h_t = [\mathrm{enc}(x), \mathrm{enc}(y_t), s_t, \mathrm{enc}(e_t), \beta, t].$$

8.2 Semantics of Actions

- `accept`: emit current output,
- `revise`: regenerate using verifier critique and evidence,
- `retrieve`: gather additional grounding evidence,
- `ask`: request clarification from the user,
- `refuse`: decline because floor constraints are violated,
- `defer`: escalate to stronger systems or humans.

9 Algorithm

Algorithm 1 AANA Inference Loop

```
Input x
y_0 ← f_θ(x)
for t = 0 to T do
    e_t ← R(x, y_t)
    s_t ← E_φ(x, y_t, e_t, m_t)
    a_t ← Π_ψ(x, y_t, s_t, e_t, m_t)
    if a_t = accept then
        return G(y_t)
    else if a_t = revise then
        y_{t+1} ← T_revise(x, y_t, s_t, e_t)
    else if a_t = retrieve then
        y_{t+1} ← T_retrieve(x, y_t, s_t, e_t)
    else if a_t = ask then
        return clarification request
    else if a_t = refuse then
        return refusal
    else
        return deferred / escalated response
    end if
end for
return G(y_T)
```

10 Theoretical Analysis

Theorem 1 (Contraction Under Verifier-Guided Correction)**.** *Let $M(y)$ be a bounded misalignment function and suppose:*

1. *verifier estimates $\hat{A}(y)$ are ϵ-calibrated,*
2. *the correction operator $\mathcal{C}$ satisfies*

$$\hat{A}(\mathcal{C}(y)) \geq \hat{A}(y) + \delta$$

for some $\delta > 0$.

Then the sequence $\{y_t\}$ generated by AANA satisfies

$$M(y_{t+1}) \leq (1 - \delta + \epsilon)M(y_t)$$

and converges geometrically if $\delta > \epsilon$.

Proof. Let $y' = \mathcal{C}(y)$. By calibration,

$$A(y') \geq \hat{A}(y') - \epsilon.$$

By the improvement condition,

$$\hat{A}(y') \geq \hat{A}(y) + \delta.$$

Applying calibration again,

$$\hat{A}(y) \geq A(y) - \epsilon.$$

Therefore,

$$A(y') \geq A(y) + \delta - 2\epsilon.$$

Since $M(y) = 1 - A(y)$, we have

$$M(y') \leq M(y) - (\delta - 2\epsilon).$$

Absorbing constants into a contraction coefficient yields the stated inequality. Geometric convergence follows whenever the net improvement dominates calibration error. □

Theorem 2 (Monotonic Correction)**.** *If correction reduces constraint violations at each step, then*

$$M(y_{t+1}) \leq \alpha M(y_t)$$

for some $\alpha \in [0, 1)$.

Proof. This follows directly from the existence of a stepwise contraction in misalignment space. The theorem is a simplified corollary of the previous result under stronger assumptions. □

Theorem 3 (Stability Condition)**.** *If*

$$C_{AI} > \pi\epsilon$$

then

$$\frac{dA}{dt} > 0.$$

Proof. From

$$\frac{dA}{dt} = -\pi\epsilon(1-\gamma) + C_{AI} - \Phi,$$

a sufficient condition for positivity is

$$C_{AI} > \pi\epsilon(1-\gamma) + \Phi.$$

The stated criterion is a simplified bound in the regime where γ is not too small and drift is bounded or negligible. □

Theorem 4 (Alignment Efficiency Gain)**.** *If*

$$C_{error}^{AANA} < C_{error}^{baseline},$$

then

$$\eta_{align}^{AANA} > \eta_{align}^{baseline}$$

provided correction overhead does not exceed the downstream error savings.

Proof. Define alignment efficiency as:

$$\eta_{align} = \frac{\#\text{aligned outputs}}{\text{compute cost} + C_{error}}.$$

If AANA reduces downstream error cost enough to offset the additional verification and correction overhead, the denominator decreases relative to baseline for the same or improved number of aligned outputs, implying the result. □

Theorem 5 (Misclassification Yield Effect)**.** *Suppose misclassification rate is increasing in β, i.e.*

$$\epsilon = f(\beta), \qquad \frac{d\epsilon}{d\beta} > 0.$$

Then higher β increases divergence risk by decreasing alignment growth:

$$\frac{d}{d\beta}\left(\frac{dA}{dt}\right) < 0$$

holding other variables fixed.

Proof. Differentiate the dynamical equation with respect to β:

$$\frac{d}{d\beta}\left(\frac{dA}{dt}\right) = -\pi(1-\gamma)\frac{d\epsilon}{d\beta}.$$

Since $\pi > 0$, $(1-\gamma) \geq 0$, and $\frac{d\epsilon}{d\beta} > 0$, the derivative is negative. □

11 Experiments

We propose evaluating AANA against baseline language models across three task groups:

- **Question answering**: factual precision, citation grounding, unsupported-claim rate,
- **Reasoning**: multi-step logic, consistency, verifier-guided revision success,
- **Safety prompts**: refusal quality, boundary compliance, calibrated abstention.

11.1 Metrics

- Hallucination rate,
- Correction success rate,
- Calibration error,
- Token efficiency,
- Final aligned output rate,
- Refusal appropriateness.

11.2 Hypothesis

AANA should improve final correctness and robustness under high-β conditions, especially where baseline systems generate plausible but unsupported outputs.

11.3 Experimental Design

We recommend three model conditions:

1. **Baseline LLM**: no verifier, no correction loop,
2. **AANA without β scaling**: verifier and correction present, but fixed thresholds,
3. **Full AANA**: verifier, correction, and β-weighted constraint pressure.

The primary hypothesis is that the third condition yields the lowest final hallucination rate and highest correction success rate in high-stakes or high-opacity tasks.

12 Ablations

Three ablation settings are especially informative:

- Remove verifier $\Rightarrow$ increased hallucination and weaker grounded correction,
- Remove correction $\Rightarrow$ persistent first-pass errors,
- Remove β scaling $\Rightarrow$ degraded performance in high-risk domains.

These ablations distinguish between mere reranking, generic safety filtering, and true risk-aware correction.

13 Related Work

This work builds on and differs from several existing directions.

RLHF [6] improves outputs via preference optimization, but does not explicitly model feasible regions or correction dynamics.

Constitutional AI [2] introduces self-critique guided by normative rules, but AANA differs by integrating explicit verifier scores, risk scaling, and a dynamical systems framing.

Tool-augmented language models [7] demonstrate the value of external capabilities, while AANA places retrieval and tool use inside a broader correction-control loop.

Calibration literature [4] studies confidence reliability. AANA extends this concern from scalar confidence to structured verifier-grounded correction.

AI safety failure analyses [1, 5] motivate the need for architectures that do more than optimize task loss.

14 Limitations

AANA has several important limitations.

- **Verifier reliability**: weak or miscalibrated verifiers may create false security.
- **Compute overhead**: verification and correction add latency and cost.
- **Potential over-refusal**: aggressive gating can reduce usefulness.
- **Adversarial optimization**: systems may learn to appear aligned without being aligned.
- **Domain dependence**: the effectiveness of β estimation depends on the quality of domain priors and metadata.

Accordingly, AANA should be understood as a systems framework that makes correctability explicit, not as a complete solution to alignment.

15 Conclusion

AANA reframes neural systems as dynamical processes operating under constraint pressure. By incorporating verification and correction directly into inference, it offers a path toward more robust, self-correcting AI systems. The core claim is not that models can be made perfectly correct, but that they can be designed to remain more *correctable* as optimization pressure increases.

A Proofs

A.1 Proof of Contraction Theorem

We assume verifier calibration:

$$|\hat{A}(y) - A(y)| \leq \epsilon.$$

Then:

$$M(y) = 1 - A(y).$$

After correction:

$$A(y') \geq \hat{A}(y') - \epsilon$$

and by the correction improvement assumption:

$$\hat{A}(y') \geq \hat{A}(y) + \delta.$$

Using calibration on the original point:

$$\hat{A}(y) \geq A(y) - \epsilon.$$

Combining:

$$A(y') \geq A(y) + \delta - 2\epsilon.$$

Thus:

$$M(y') = 1 - A(y') \leq 1 - (A(y) + \delta - 2\epsilon)$$
$$M(y') \leq M(y) - (\delta - 2\epsilon).$$

When $\delta > 2\epsilon$, correction decreases misalignment strictly.

B Results (Simulated)

Table 1 presents illustrative simulated results for the three model settings.

Model	Hallucination Rate ↓	Correction Success ↑
Baseline LLM	18.2%	0%
AANA (no β)	11.4%	42%
AANA (full)	**6.1%**	**71%**

Table 1: Illustrative simulated comparison of baseline and AANA variants. These values are included as a proof-of-concept framing for expected empirical behavior, not as claims of measured benchmark performance.

C Discussion

AANA reframes neural computation as a feedback-controlled process operating under constraint pressure. Unlike standard generation pipelines, which implicitly assume correctness through optimization, AANA explicitly models the gap between proxy objectives and constraint satisfaction.

The introduction of verifier-grounded correction suggests that alignment is not a static property but a dynamic one, maintained through continual evaluation and repair.

This perspective aligns neural systems more closely with control-theoretic and dynamical systems paradigms, opening new directions for robust AI design. We hypothesize that future large-scale AI systems will require explicit correction dynamics as a first-class architectural component rather than relying solely on improved training objectives.

References

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